ACS NANO

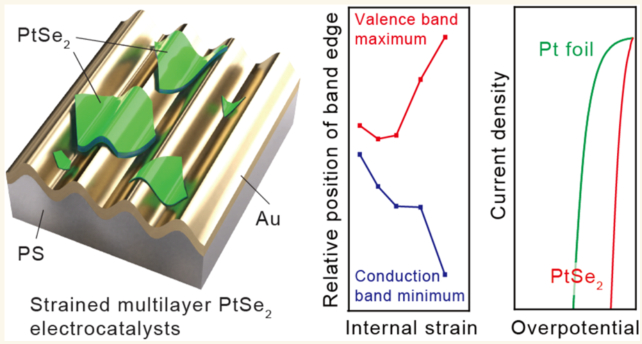
www.acsnano.org

Strain-Enabled Band Structure Engineering in Layered $PtSe_2$ for Water Electrolysis under Ultralow Overpotential

Hotae Jeon,○ Hee Jung Kwon,○ Jaehyun Lee,○ Sun Kyung Han, Hyunjin Kim, Jaewon Heo, Junhwi Han, Seunghun Shin, Jiheon Park, Min Kyung Cho, Daniel J. Preston, In Soo Kim,* Minho Kim,* and Won-Kyu Lee*

Cite This: *ACS Nano* 2025, 19, 9107−9120

ABSTRACT: This paper describes a simple design methodology to develop layered $PtSe_2$ catalysts for hydrogen evolution reaction (HER) in water electrolysis operating under ultralow overpotentials. This approach relies on the transfer of mechanically exfoliated $PtSe_2$ flakes to gold thin films on prestrained thermoplastic substrates. By relieving the prestrain, a tunable level of uniaxial internal compressive and tensile strain is developed in the flakes as a result of spontaneously formed surface wrinkles, giving rise to band structure modulations with overlapped values of the valence band maximum and conduction band minimum. This strain-engineered $PtSe_2$ with an optimized level of internal tensile strain amplifies the HER performance of the $PtSe_2$, with performance far greater than that of pure platinum due to significantly reduced charge transfer resistance. Density functional theory calculations provide fundamental insight into how strain-induced band structure engineering correlates with the promoted HER activity, especially at the atomic edge sites of the materials.



KEYWORDS: *water electrolysis, hydrogen evolution reaction, catalytic materials, transition metal chalcogenides, strain engineering, electrochemical processes*

INTRODUCTION

Platinum diselenide ($PtSe_2$), a group-10 transition metal dichalcogenide (TMD), has exhibited promise for applications in nanoelectronics,[1−7] optoelectronics,[8−12] and spintronics[13−15] owing to its ultrahigh carrier mobility, stability in air, and unique layer-dependent band gaps.[1,2,16] Theoretical calculations and experiments have verified that $PtSe_2$ undergoes a semiconductor-to-metal transition as its thickness increases from monolayer to trilayer.[16] Layered (semi-)metallic $PtSe_2$ is of particular importance for water electrolysis because undercoordinated Pt atoms at the edge sites are highly (re)active for electrocatalytic hydrogen evolution reaction (HER).[17] Because the basal planes of mono- and multilayer $PtSe_2$ lack (re)active sites for hydrogen adsorption and reduction, various approaches—for example, plasma treatments for defect engineering (e.g., formation of Se vacancies) and amorphization,[18] electrochemical reduction to form Pt/$PtSe_2$ heterointerfaces,[19] and templated growth to produce vertical nanosheets[20]—have improved overall HER performance from this material system. These methods, however, typically require expensive vacuum equipment and multistep synthetic processes, limiting the scalability and cost-effectiveness of the electrocatalysts. Furthermore, although some engineered $PtSe_2$ catalysts have exhibited HER performance greater than pure Pt (with fabrication based on the complex multistep processes described above),[17,18] the reported HER performance of most other Pt-based TMDs is typically still considerably lower than that of Pt.

Straining two-dimensional (2D) TMDs represents a practical approach to modulating mechanical, (opto-)electrical, and thermoelectric properties of the materials over large areas

Received: December 13, 2024
Revised: February 18, 2025
Accepted: February 20, 2025
Published: February 26, 2025

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

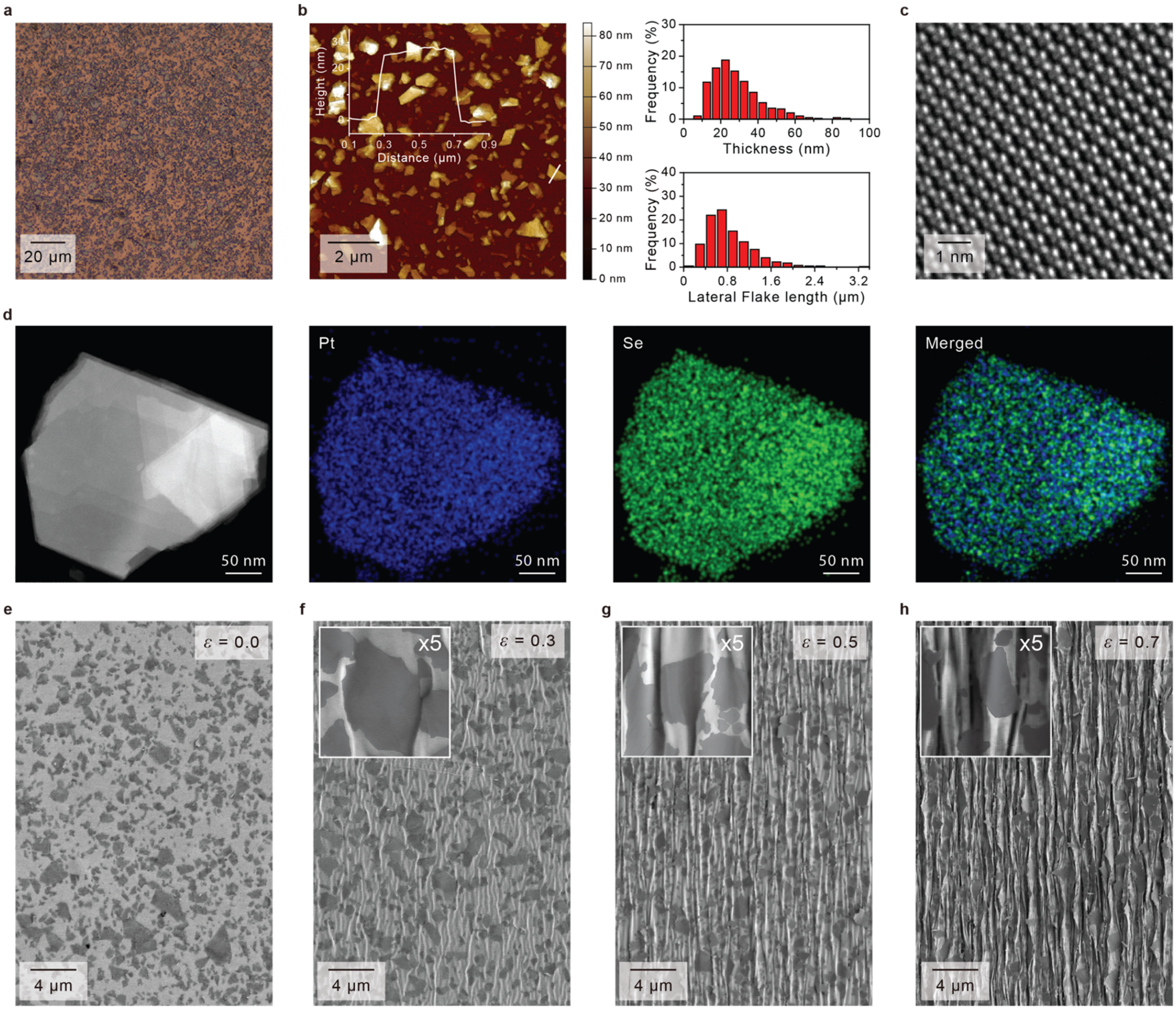


Figure 1. Formation of strain-engineered bulk $PtSe_2$ by surface buckling. (a) OM- and (b) AFM image of as-transferred $PtSe_2$ flakes (left) and thickness and size distributions of the flakes characterized by averaging the lengths of individual flakes using their cross-sectional profiles (right). The inset shows the cross-sectional profile of the AFM image of the $PtSe_2$ corresponding to the white line, depicting characteristic thickness. (c) HRTEM image and (d) compositional maps of the $PtSe_2$ flake. SEM images of $PtSe_2$ before (e) and after straining at (f) ε = 0.3, (g) ε = 0.5, and (h) ε = 0.7. For the data shown in (b), we characterized ∼400 flakes.

(>cm^2) without relying on complex equipment or costly infrastructure.[21−26] Surface wrinkling of TMDs on elastomeric substrates represents a massively parallel tool to pattern strain distributions; tensile strains up to ∼2.5% have been developed over the crest areas of wrinkled MoS_2, resulting in spatially tunable optoelectronic properties with modulated bandgaps.[27,28] In particular, local semiconductor-to-metal phase transitions have been reported in mechanically exfoliated MoS_2 and $MoTe_2$ by wrinkling the layered flakes on viscoelastic substrates, resulting in substantially enhanced HER with electrochemical activation.[29,30] These results suggest a pathway to amplify the electrocatalytic reactivity and HER activity of $PtSe_2$: strain engineering of the type-II Dirac semimetal controls its band structures and surface and interlayer charge density/distributions, and it has been calculated that the monolayer and bilayer $PtSe_2$ show reduced band gaps under either compressive or tensile strain.[31] These modulated electronic structures of Pt-based TMDs enhance the required interfacial charge transfer such that scalable, cost-effective, and ultraefficient water splitting with performance much greater than that of Pt alone becomes possible.

RESULTS AND DISCUSSION

In this work, we demonstrate for the first time that HER from multilayer (or bulk) $PtSe_2$ can be optimized by leveraging internal compressive- and tensile strains with modulated band structures. Direct dry transfer of mechanically exfoliated $PtSe_2$ flakes to thin gold films deposited on prestrained thermoplastic substrates provides a platform for producing scalable working electrodes for HER. By relieving the prestrain macroscopically in a tunable manner, uniaxial tensile strains (up to +8%) and compressive strains (down to −6%) developed in the flakes along with spontaneously formed microwrinkles, verified using Raman spectroscopy. Both X-ray and ultraviolet photoelectron

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

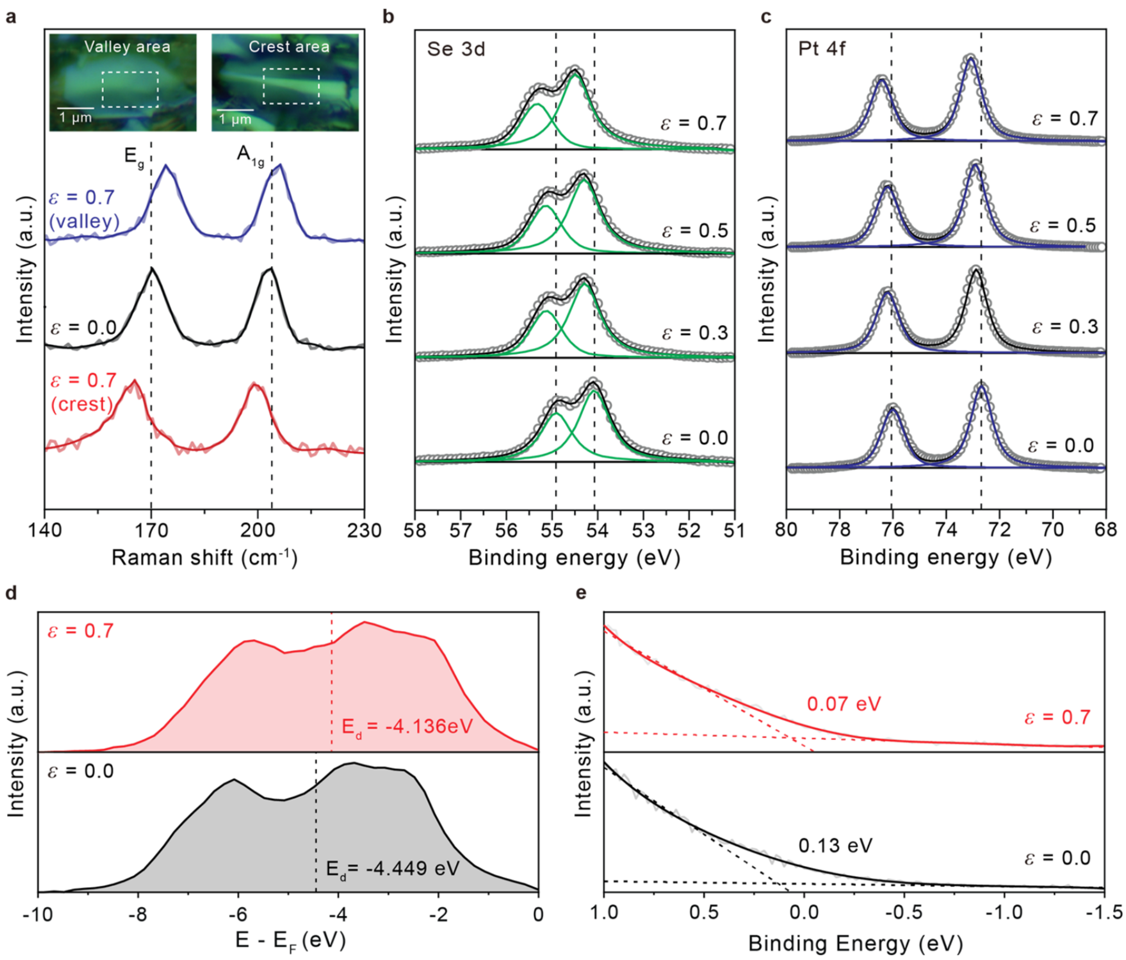


Figure 2. Spectroscopic understanding of strain-engineered bulk $PtSe_2$. (a) Raman spectra of as-transferred and strained $PtSe_2$ at $\varepsilon = 0.7$ over local areas of valley and crest of a single buckle. XPS Se 3d (b) and Pt 4f (c) peaks of the $PtSe_2$ before and after buckling at different strains. (d) d-band centers of as-transferred and the strained $PtSe_2$ measured by VB-XPS, and (e) energy difference ($\Delta E = E_f - E_{VBM}$) values estimated from UPS measurements to understand surface electronic structures of $PtSe_2$ before and after straining.

spectroscopy analysis suggest that this strain engineering approach modulated the surface electronic structures of $PtSe_2$ such that the resulting electron density and distributions on the catalysts benefitted HER. Theoretical calculations show that the strain-enabled band structure engineering in layered $PtSe_2$, with tunable overlapping between the conduction band minimum and the valence band maximum, supported nearly thermoneutral hydrogen adsorption on the Pt and Se edge sites such that HER greater than that of Pt alone resulted, which we demonstrated with detailed experimental measurements.

We mechanically exfoliated multilayer $PtSe_2$ from mineral crystals and then transferred the nano and microscale flakes to gold (Au) thin films. These thin films were deposited on prestrained thermoplastic polystyrene (PS) following hot pressing (Figures 1a and S1 and Methods section). Statistical analysis of the thickness and size distributions of the as-transferred $PtSe_2$ showed that the flakes we transferred onto the Au surfaces had an average thickness of ~28 nm (i.e., ~54 layers of $PtSe_2$) and a lateral length of ~800 nm (Figure 1b). The high-resolution TEM (HRTEM) image showed the atomic fringes of the layered flakes, indicating a crystalline 1T phase structure, and the energy dispersive X-ray (EDX) maps verified the presence of stoichiometric $PtSe_2$ (Figures 1c,d and S2). We uniaxially relieved the prestrain by heating the substrates above the PS glass transition temperature (T_g) to generate ordered Au microwrinkles with buckled $PtSe_2$, where the amount of macroscopic strain (ε) is tunable by varying the heating time (Figure S1 and Methods section) as well as the level of prestrain. Here, we mechanically clamped each end of the substrates before heating to induce the uniaxial strains. Wrinkles formed across the entire surface of the centimeter-scale sample in parallel because of the difference between Young's modulus of the skin layer (in this case, the Au film partially covered with $PtSe_2$) and the substrate.[32−36] The scanning electron microscopy (SEM) images showed the flat surfaces of Au films with $PtSe_2$ flakes transformed to three-dimensional (3D) microwrinkles after straining such that flakes with a lateral size larger than a few hundred nanometers buckled out-of-plane (Figure 1e−h). As ε increased, the buckled flakes were confined more tightly between adjacent Au wrinkles because the wrinkling wavelength decreased with strain.[29,30]

In order to understand the changes in the electronic structure of the $PtSe_2$ upon applied strain at the single flake level, we performed Raman spectroscopy (Figure 2a). The Raman spectrum (solid black line) of the as-transferred sample exhibits two discrete phonon modes representative of $PtSe_2$; E_{2g} at 170 cm^{-1} and A_{1g} at 203 cm^{-1} associated with lateral and vertical vibrations of the $PtSe_2$ lattice, respectively.[37−39] The formation of buckles upon applied macroscopic strain (ε of

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

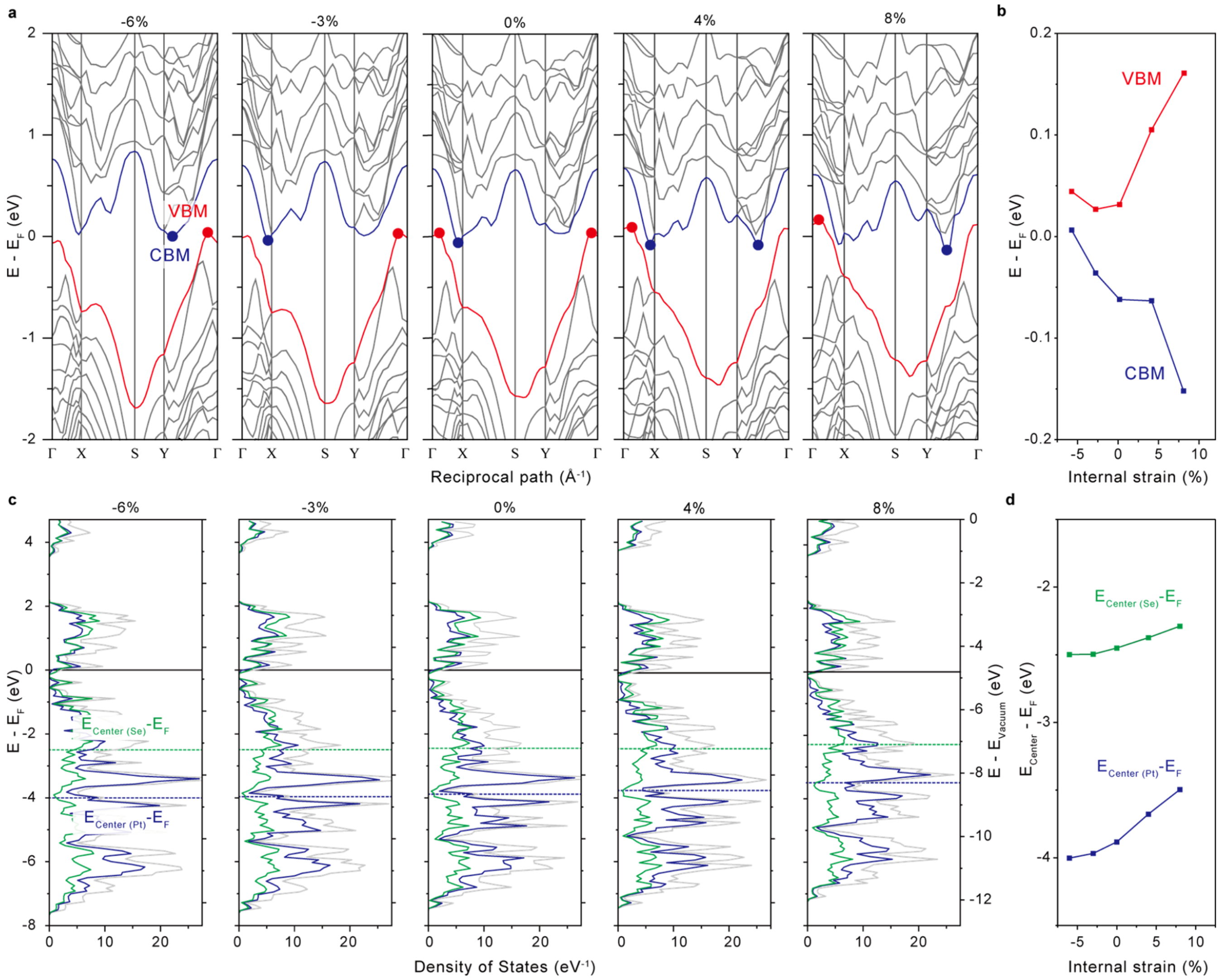


Figure 3. Theoretical calculations on band-structure variations of strain-engineered layered $PtSe_2$. (a) Band structures of trilayer $PtSe_2$ before and after strain engineering and (b) the relative positions of VBM and CBM with respect to E_f. (c) Corresponding total density of states (DOS) and projected DOS of Pt and Se. (d) The relative band center positions of Pt and Se as a function of applied internal strain.

0.7) spontaneously gives rise to "valley" regions with compressive strains and "crest" regions with tensile strains.[27,28] Shifts in both phonon modes occurred based on the nature of strain by virtue of 3D wrinkling, where the crest (valley) region exhibits a shift to lower (higher) wavenumbers by ~4.5 cm^{-1} (~3.4 cm^{-1}), suggesting internal tensile (compressive) strain values of 8.7% (6.6%) developed in the flakes of $PtSe_2$.[40,41] The level of strain developed in the layered $PtSe_2$ was quantitatively comparable to the cases of multilayer MoS_2 and $MoTe_2$ strained with the same material processing technique.[29,30] Furthermore, the level of the Raman shift was preserved after HER, indicating that the strained lattices of $PtSe_2$ were not affected by water-based electrolytes to benefit the long-term stability of the catalysts as we show in the section on HER characterization (Figure S3).

While Raman shifts by straining have been considered as direct signs of doping to control the band structures of the type-II Dirac semimetal,[42,43] we further performed X-ray photoelectron spectroscopy (XPS) because the coexistence of both red- and blue-shifts complicates the analysis of the overall strain effects on the layered $PtSe_2$. In contrast to Raman spectroscopy, techniques such as XPS and ultraviolet photoelectron spectroscopy (UPS) statistically analyze the collective variations of electronic structures because the large beam spot diameter covers multiple flakes on the samples (Methods section). Both Pt and Se core level features of the as-transferred $PtSe_2$ illustrated in Figure 2b,2c are consistent with previous reports in the literature.[44−46] Specifically, the binding energies located at 54.1, 54.9, 72.7, and 76.0 eV are associated with Se $3d_{5/2}$, Se $3d_{3/2}$, Pt $4f_{7/2}$, and Pt $4f_{5/2}$, respectively. Upon application of strain, the binding energies of both Pt and Se core levels shift to higher energies with increasing amounts of strain. Here, the positive shift in binding energies indicates electron doping of $PtSe_2$, which will lift the Fermi level toward the conduction band minimum (CBM).[42] We verified this behavior using valence band spectra obtained by XPS (VB-XPS, Figure 2d and Methods section); the d band center (E_d) of $PtSe_2$ was increased by ~0.31 eV with ε of 0.7, suggesting that the density of states moves closer to the Fermi level (E_f) with macroscopic straining, which potentially benefits surface electrochemical reactions as we show in the following sections regarding theoretical free energy values for hydrogen adsorptions.

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

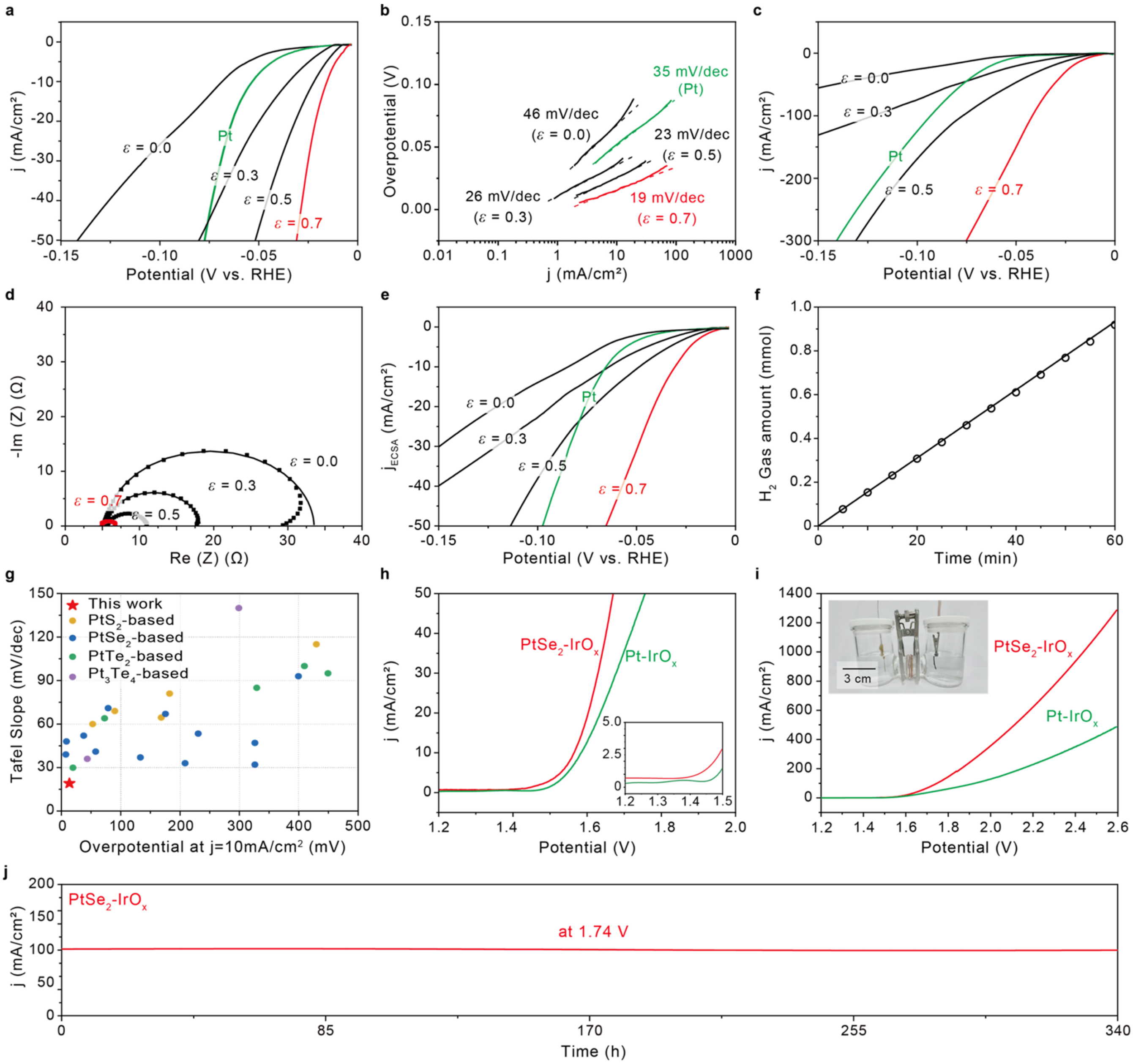


Figure 4. Electrocatalytic performance of strain-engineered bulk $PtSe_2$. (a) iR-corrected polarization curves from as-transferred (ε = 0.0) and strained (ε = 0.3, ε = 0.5, and ε = 0.7) $PtSe_2$ compared to results obtained for Pt, and (b) corresponding Tafel plots obtained from the curves. (c) iR-corrected polarization curves of the EC cells in (a) over the high current density regime. (d) Nyquist plots of the EC cells with $PtSe_2$ before and after straining at different ε. (e) Polarization curves normalized to the C_{DL} estimated in Figure S12. (f) Comparison between the amount of collected and theoretical gaseous product (H_2) by the electrochemical cell containing strained $PtSe_2$ at ε = 0.7 (as a working electrode) at a constant current density of 100 mA/cm² in 0.5 M sulfuric acid electrolyte at 25 °C. (g) HER performance of strain-engineered $PtSe_2$ at ε = 0.7 in comparison to (Pt-based) transition metal chalcogenides electrocatalysts published in literature. *I*−*V* curves of overall water splitting in the electrolyzer over (h) low- and (i) high current density regime. The electrolyzer containing Pt foil as a cathode was used as a comparison group in (h, i). (j) The stability of the water splitting from the $PtSe_2$-based electrolyzer using chronoamperometry at 1.74 V. The inset in (i) indicates photograph of the two-electrode electrolyzer consisting of an IrO_x anode (in alkaline electrolyte), a $PtSe_2$ cathode strained at ε = 0.7 (in acidic electrolyte), and a bipolar membrane as also depicted in Figure S15.

We also performed UPS measurements to observe the behavior of strain-induced electronic structure transformation (Methods section), where (i) the cutoff energy in the valence band spectrum provides the position of valence band maximum (VBM) from the Fermi level ($\Delta E = E_f - E_{VBM}$) and (ii) the secondary cutoff spectrum provides work function values based on the difference from the energy of the He I excitation source (21.22 eV).[47,48] As illustrated in Figure 2e, we confirmed that E_{VBM} increases by ~0.06 eV upon application of strain, which verifies an upshift in the Fermi level as discussed in the XPS results corroborating the effects of electron doping and the upshift of E_d in strained $PtSe_2$. In addition, the work functions of −4.14 and −4.68 eV were determined for as-transferred and strained $PtSe_2$, respectively, which indicates an increased work function in strained $PtSe_2$ (Figure S4). We note that the tensile strains in TMDs have

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

been theoretically estimated to change the vacuum levels of the materials, partially attributing to the varying work function in this material system.[49]

We theoretically explored the band structures of the layered $PtSe_2$ before and after strain engineering based on density function theory (DFT) calculations (Figure 3). Here, the internal uniaxial lattice strains between −6% (i.e., compressive strain) and 8% (i.e., tensile strain), developed by the macroscopic ε and determined by the amounts of Raman red- and blue shifts, were applied along the model system of trilayer $PtSe_2$ (Methods section). The semiconductor-to-metal transitions occur in multilayer $PtSe_2$ with overlaps between E_{VBM} and E_{CBM} as the layer number increases; in this work, the strain-free model (with the unit cell of $\Delta a/a_0 = 0\%$) exhibited a higher energy level of the VBM ($E_{VBM} - E_f = +0.03$ eV) than that of the CBM ($E_{CBM} - E_f = -0.06$ eV), matching with the literature (Figure 3a).[16,17] In comparison with the $PtSe_2$ with zero strain, tensile strains ($\Delta a/a_0 = +4$ and +8%) tend to raise the E_{VBM} ($E_{VBM} - E_f = +0.10$ and +0.16 eV, respectively) and lower the E_{CBM} ($E_{CBM} - E_f = -0.06$ and −0.15 eV, respectively), which maximizes the band overlaps (Figure 3b). In contrast, the compressive strain down to −6% increased both E_{VBM} ($E_{VBM} - E_f = +0.04$ eV) and E_{CBM} ($E_{CBM} - E_f = +0.01$ eV), resulting in a slightly reduced band overlap. Considering the averaged increment of E_{VBM} by ∼0.06 eV measured by UPS which covers numerous flakes (Figure 2e), both local tensile (from crest regions of a buckle) and compressive (from valley regions of a buckle) strains collectively contributed to the reconstructed valence and conduction bands overlapping with each other.

We calculated the density of states (DOS) of the layered $PtSe_2$ to further highlight a gradual upshift of the valence band centers of Pt and Se with applied tensile strains (Figure 3c). As the internal strain changes from compression (−6%) to tension (+8%), the band center of Pt (and Se) with respect to E_f increased from −4.00 eV (−2.50 eV) to −3.50 eV (−2.29 eV) (Figure 3d), similar trend with reported multimetallic materials.[50] Considering that the relative band center of Pt (and Se) before straining was initially positioned at −3.88 eV (−2.45 eV), this result theoretically confirms that the tensile strain (rather than compressive strain) is the dominant design parameter to control the electronic structures of layered $PtSe_2$ as already verified by VB-XPS with increased E_d (Figure 2d). Because the position of the band center with respect to E_f has a strong correlation with hydrogen adsorption at Volmer step, our strategy of strain-enabled band structure engineering rationally optimizes the catalytic reactivity and activity of layered $PtSe_2$.[51,52]

To quantify how much the strain-induced band-structure engineering of $PtSe_2$ enhanced HER activity (Figure 4), we connected copper (Cu) wires with the Au wrinkles supporting structured $PtSe_2$ to fabricate working electrodes (WE) for HER in a three-electrode configuration containing a Pt foil counter electrode (CE) and an Ag/AgCl reference electrode (RE) (Figure S5 and Methods section). HER measurements with $PtSe_2$ on Au electrodes before and after straining were performed using the three-electrode cell with a 0.5 M sulfuric acid electrolyte. From the polarization curves for $PtSe_2$, we observed that the as-transferred flakes are highly active with an onset overpotential (at low current density, j, of 2 mA/cm^2) of −0.038 V vs the reversible hydrogen electrode (RHE), for which HER performance is far greater than the Au films without $PtSe_2$ (Figures 4a and S6). In general, the HER efficiency of TMDs decreased with increasing thickness because interlayer charge transfer kinetics and the ratio of active edge sites to the entire volume of catalysts were reduced.[53] In contrast, it has been demonstrated that the HER activity of $PtSe_2$ increases with the number of layers due to the enhanced interlayer coupling and charge distributions, in agreement with our experimental observation in which the thick flakes consist of ∼50 layers on average.[17,54]

We observed that the HER performance of layered $PtSe_2$ enhanced as ε increased to 0.7; the overpotential at a high current density (j of 10 mA/cm^2) was suppressed from −0.069 (before straining) to −0.015 (V vs RHE) with ε, suggesting that substantially improved electrocatalytic activity of layered $PtSe_2$ was achieved by straining (Figure 4a). Considering that the potential required to achieve the same j with Pt was measured as −0.053 (V vs RHE), ∼3.5 times higher than that for the $PtSe_2$ (strained at ε = 0.7), our approach enables efficient HER operating over ultralow overpotential regimes. We characterized the HER kinetics of $PtSe_2$ modified with ε by plotting Tafel slopes based on the corresponding polarization curves (Figure 4b). The Tafel slope of as-transferred flakes (i.e., ε of 0.0) was ∼46 mV/dec, indicating that both the Volmer (Tafel slope of ∼120 mV/dec) and Heyrovsky (Tafel slope of ∼40 mV/dec) reactions could possibly be the rate-determining step.[55] The reaction kinetics of $PtSe_2$ changed after straining at ε of 0.7, with reduced Tafel slopes of ∼19 mV/dec (much lower than ∼35 mV/dec of Pt), which suggests that the Tafel reaction mechanism is responsible for HER and the desorption of hydrogen is the rate-limiting step.[56] We confirmed that the improved HER with the reaction mechanism is valid for the polarization curve with the corresponding Tafel slope (26 mV/dec) without iR correction (Figure S7). We further characterized the HER of the same samples over much higher current density regimes (Figure 4c); the $PtSe_2$ strained at ε = 0.7 only requires an overpotential of −0.075 V for j of 300 mA/cm^2, whereas Pt consumed the larger potential of −0.140 V to generate the same j value.

To understand the improved catalytic activity resulting from strain engineering, we performed electrochemical impedance spectroscopy (EIS) analysis for the electrochemical cells containing the strain-engineered $PtSe_2$ (Figures 4d and S8 and Methods section). The charge transfer resistance (R_{CT}) determined by Nyquist plots of all the samples decreased significantly as ε increased, revealing that the electron transfer between the $PtSe_2$ and the protons in the electrolyte solution increased and resulted in the reduced Tafel slopes. We observed that the decreases in R_{CT} were partially attributable to substrate structuring effects with improved surface wettability and mass transport for HER (Figures S9 and S10). We note that the serial resistance of the electrochemical cells containing $PtSe_2$ working electrodes was nearly identical before and after strain engineering (Figure S8). Furthermore, we quantified the HER performance for $PtSe_2$ under the maximum strain achievable with the Au/PS wrinkling system (ε of 0.8) and found that it was slightly lower than that from the sample with ε of 0.7 due to the increased level of R_{CT} attributed by the formation of microcracks on Au wrinkles (Figure S11).

We estimated the electrochemically active surface area (ECSA) for each cell based on the electrochemical double-layer capacitance (C_{DL}), the value of which was determined by measuring the non-Faradaic capacitive current associated with double-layer charging from the scan-rate dependence of cyclic voltammetry (CV) (Figure S12 and Methods section). The

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

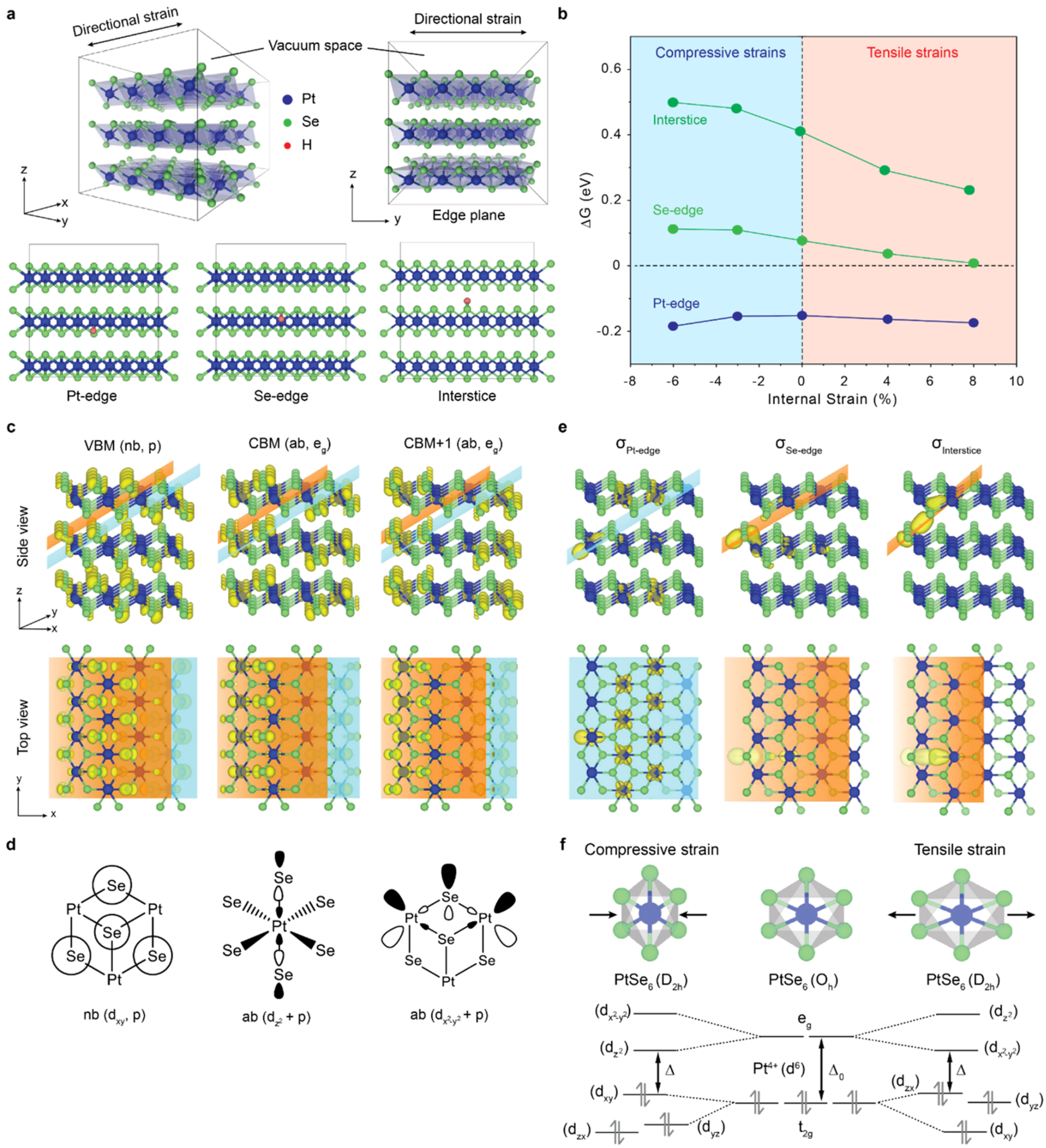


Figure 5. Theoretical understanding of electrochemical hydrogen evolution from strain-engineered, layered $PtSe_2$. (a) Atomic model structures of the edge plane of trilayer $PtSe_2$ slab with various H adsorption sites. The compressive and tensile strains were applied along the y direction. (b) The adsorption energy of hydrogen atom at the 1st Volmer step (ΔG_H in eV) with strain. ΔG_H was measured at edge Pt and Se atomic sites and interstitial adsorption sites.[17] (c) Band-decomposed charge densities of the VBM, CBM and CBM + 1 in the model structure with the parallel alignment planes containing the bonding axes of Se−H and Pt−H in orange and cyan, respectively, and (d) schematic orbital diagrams of VBM (nonbonding, nb, of Pt d_{xy} and Se p), CBM (antibonding, ab, of Pt d_{z^2} and Se p) and CBM+1 (ab of Pt $d_{x^2-y^2}$ and Se p). (e) Band decomposed charge density of the bonding states of the H atoms in the Volmer intermediates. (f) The d orbital splitting of a $PtSe_6$ octahedron with uniaxial compressive and tensile strains. The crystal field splitting parameters of octahedral (O_h) and distorted $PtSe_6$ (D_{2h}) are shown in Δ_o and Δ, respectively.

C_{DL} of as-transferred $PtSe_2$ was monotonically enhanced by ~260% (from 64.6 to 170.2 $\mu F/cm^2$) with ε increasing to 0.7. By assuming that C_{DL} is proportional to ECSA,[57,58] we confirmed that the strain-enabled band structure engineering amplified the density of catalytically active sites for hydrogen evolution. Figure 4e shows the polarization curves normalized

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

to the C_{DL} of the corresponding cells, and the relative configurations of the normalized curves (i.e., the current density per active area as a function of overpotential) matched with the current density (per geometric area) over the same overpotential range in Figure 4a. The exchange current density (j_0), which is the relevant thermodynamic parameter for the HER rate at zero overpotential, scales with ε, suggesting that the band structure engineering enhancing ECSA also benefits the steady-state catalytic activity in overcoming Pt (Figure S13). We evaluated the performance stability of strained $PtSe_2$ (at ε of 0.7) using chronoamperometry and confirmed that the resulting HER was much more stable than that of Pt (Figure S14). The Faradaic efficiency of the HER from the sample was verified at ∼98% (Figure 4f and Methods section).

Figure 4g compares HER performance of strain-engineered $PtSe_2$ in this work with reported Pt-based TMD electrocatalysts (summarized in Table S1). The amorphization of monolayer $PtSe_2$ demonstrated state-of-the-art performance (Tafel slope of 39 mV/dec, ∼0.025 V for j = 10 mA/cm^2), comparable in efficiency with the HER in this work.[18] The fabrication for these amorphous catalysts, however, contains multistep selenization and crystallization processes over varying temperatures followed by deep reactive ion etching (at a low temperature of −30 °C) that would limit the cost-effectiveness of the materials system. Although ordered clustering of Te vacancies in $PtTe_2$ nanosheets exhibited one of the most efficient HER (Tafel slope of 29.9 mV/dec and 0.022 V for j = 10 mA/cm^2), the materials processing requires electrochemical exfoliation of bulk crystals and the collection of the exfoliated products by centrifugation (which follows delicate thermal treatments in a tube furnace after multistep rinsing and drying).[59] Considering the simple fabrication processes, consisting only of mechanical exfoliation and strain engineering, as well as considerable electrocatalytic activity, we believe our system has potential to lead to a rational green-hydrogen production scheme.

As a proof-of-concept demonstration, we constructed a lab scale two-electrode electrolyzer cell based on strain-engineered $PtSe_2$ for water splitting (Figures 4h, S15, and S16). This full cell consists of the strain-engineered $PtSe_2$ (at ε of 0.7) as a cathode and an IrO_x on carbon cloth (CC) as an anode (noted as $PtSe_2$−IrO_x cell). The cell exhibited an onset potential of 1.47 V for the reaction, lower than the potential of 1.52 V for Pt-IrO_x cell (a comparison electrolyzer consisting of Pt foil as a cathode and IrO_x/CC as an anode), which implies that the strained $PtSe_2$ operating under ultralow overpotential for HER indeed benefits the overall water splitting process. We highlight that the performance gap between these two cells increased significantly over high current density regimes; the $PtSe_2$−IrO_x cell generates j up to ∼1300 mA/cm^2 at a potential of 2.6 V, whereas only 38% of the current level (i.e., ∼500 mA/cm^2) was induced from the Pt-IrO_x cell at the same potential (Figure 4i). This level of j over 1000 mA/cm^2 suggests that industrial applications of the $PtSe_2$-based electrolyzer for mass production of molecular hydrogen are potentially attainable.[18] We also note the HER from the reference Pt-IrO_x cell over low- and high-current regimes is similar to the previous reports.[60] We further characterized the long-term stability of the $PtSe_2$−IrO_x cell with chronoamperometry; the electrolyzer maintained j of 100 mA/cm^2 at the overpotential of 1.74 V for 340 h without performance drops (Figure 4j). We confirmed that this excellent stability of the cell is also valid for lower overpotential regimes with j of 10 and 50 mA/cm^2, which indicates the compressive and tensile strains in the $PtSe_2$ are preserved in aqueous electrolytes with applied overpotentials (Figure S17). Considering that the HER from strained $PtSe_2$ is preserved in proton exchange membranes (PEMs), our system has a potential as a core component of a PEM electrolyzer (Figure S18); we checked that the formation of buckled flakes on a PEM is possible using nanoimprinting processes, giving rise to a suitable platform for PEM water electrolysis (Figure S19).

We mechanistically investigated how the strain engineering of multilayer $PtSe_2$ affected the resulting electrochemical (re)activities of the catalysts at the atomic scale (Figure 5). In general, the edge planes of layered $PtSe_2$ contain highly reactive, unsaturated Pt (Pt-edge) and Se atoms for hydrogen adsorptions such that the larger number of layers resulted in greater HER at the single flake level.[17] We note that the Se sites adsorb an H atom along the edge direction (Se-edge) and interstitial direction between adjacent layers (interstice) (Figure 5a). Figure 5b shows the change in Gibbs energy (ΔG_H) during the Volmer step of initial hydrogen adsorption on Pt and Se edge atoms calculated using DFT, in which we found both Se-edge and interstitial sites are strain sensitive. As the applied strain varies from compression (−6%) to tension (+8%), the ΔG_H of interstice decreased from ∼+0.50 to ∼+0.22 eV, where HER reactivity similar to Pt(111) is possible (Figure S20).[61,62] We highlight that the Se-edge with the tensile strain (+8%) induced nearly thermoneutral hydrogen adsorption with ΔG_H of ∼+0.01 eV, which results in more favorable H_2 evolution than Pt(111) with slightly stronger H adsorption; ΔG_H of ∼−0.24 eV was calculated from our Pt(111) model with a comparable surface area of the edge-plane model of layered $PtSe_2$ (Methods section and Figure S20). While a hard-acid Pt^{4+} provides a relatively strong adsorption site for a hard-base H atom (•H), a soft-base Se^{2-} induces a suitably reversible adsorption site for a hard-acid H cation (H^+) that can be reductively adsorbed to form a Volmer intermediate. Although the hydrogen adsorption on the Pt-edge was nearly strain-insensitive over the same strain range, the average ΔG_H of ∼−0.016 eV from the atomic sites theoretically supports the experimental HER greater than Pt.

The charge density of band edges, VBM, CBM, and CBM + 1 (the energy level right above CBM) illustrates the alignment planes of Pt d and Se p orbitals near the Fermi level in the layered $PtSe_2$ (Figure 5c). The VBM mostly consists of nonbonding (nb) Se 4p states attributed to the Se^{2-} anionic states, while the CBM and CBM + 1 are composed of the antibonding (ab) Pt d and Se p orbitals (Figure 5d), in accordance with previous literature.[17] With the adsorption of H atoms, both Pt d_{z^2} and Se p_z at the edge sites construct bonding orbitals (σ) with H 1s orbitals (Figure 5e), mostly attributed to the CBM states of $PtSe_2$. The interstitial site generates the interlayer coupling between the adjacent layers and the adsorbed H atom on the site stabilizes the Volmer intermediate. We checked that the dihedral angle (φ) of layered $PtSe_2$ before and after H atom adsorption at the Pt (and Se-edge) site was similar whereas φ decreased substantially when the Volmer intermediate formed at the interstitial site (Figure S21). This large variation of φ before and after hydrogen adsorption indicates that the interstitial adsorption involves steric hindrance and electronic repulsion between the H atom and the $PtSe_2$ layers, resulting in a thermodynamically less-favored ΔG_H among the edge sites.

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

As we apply the uniaxial tensile strains to layered $PtSe_2$, the $PtSe_6$ octahedron as a unit cell undergoes the Jahn−Teller distortion to strengthen the splitting of the CBM states such that the resulting CBM state becomes more stabilized from the CBM + 1 state (Figure 5f).[63] This CBM state closer to the Fermi level increases the n-type carrier concentration (as observed with XPS analysis) and promotes the preferential reductive adsorption of an H atom.[64] As the internal strain increases, the reduced crystal field splitting parameter (Δ) results, and the strain-induced variations in the energy levels of electrons in layered $PtSe_2$ were indicated by the strain-enabled CBM-VBM overlapping (Figure 3a,b).[65] Since uniaxially applied strain is parallel with the xy plane and perpendicular to the z axis in the model (Figure 5a), the energy level of d orbitals aligned with the z axis (i.e., d_{z^2}, d_{yz}, and d_{zx} orbitals) are raised by a tensile strain and lowered by a compressive strain, whereas the in-plane d orbitals (i.e., $d_{x^2-y^2}$ and d_{xy} orbitals) vary with the strains vice versa (Figure 5f). Therefore, both of compressive and tensile strains reduce the overall d splitting energy from the octahedral splitting energy. Since the VBM and CBM positions of $PtSe_2$ are determined by the positions of the filled- and empty d orbitals, respectively, the resulting E_{VBM} (E_{CBM}) tend to increase (decrease) by either compressive- or tensile strains (Figure S22a).

The tensile strains not only induce the distortion of the $PtSe_6$ octahedron, but also affect the interlayer coupling which can facilitate the HER activity.[54] As the level of tensile strain increases along the zigzag atomic arrangements of $PtSe_2$ (i.e., the direction parallel to y axis in Figure 5a), the CBM states (d_{z^2}) of the adjacent layers contribute to interlayer electronic coupling that can further stabilize the Volmer intermediates at the Se-edge sites (Figure S23). Because the zigzag tensile strain elongates Pt−Se bonds along the basal plane of $PtSe_2$, the Pt d_{z^2} (interacting weakly with the bonded Se atoms) strongly involves in the interlayer coupling and can be localized. We also checked that as the strain varies from compression to tension, the Se p orbitals are decoupled from the valence doubly degenerated, nonbonding Pt d states and strengthen the interlayer p−p coupling for favorable H adsorption at the Se edge sites (Figures S22b,c). Therefore, we highlight that a transition of thermodynamic preference from intralayer coupling to the interlayer coupling arises as the uniaxial tensile strain develops in the layered system; this interlayer coupling further enhances the hydrogen adsorptions on the edge sites by stabilizing alignments between the Se−H bonding orbitals and Pt d_{z^2} in the adjacent layer.

CONCLUSIONS

In summary, we developed an approach to modulate the electrocatalytic activity of layered $PtSe_2$ through strain-induced band structure engineering in a programmable manner. By optimizing the macroscopic strain relief of the thermoplastic substrates supporting $PtSe_2$ to obtain maximum overlap between the CBM and VBM of the material, we realized HER catalyst performance greater than Pt over centimeter-scale areas. DFT calculations show that the Pt and Se atoms at the edge planes induced nearly thermoneutral hydrogen adsorption under internal tensile strains, giving rise to the superior per-site HER (re)activity. Considering that strain-engineered Mo-based TMDs (e.g., MoS_2 and $MoTe_2$) with heterophase boundaries require poststructuring electrochemical processes for their optimized HER,[29,30] the ultraefficient HER from $PtSe_2$ utilizing only a single-step straining procedure is novel and practically useful for water electrolysis in terms of system operation. Because this mechanical process can easily be applied to any exfoliated transition metal chalcogenide (beyond MoS_2, $MoTe_2$, and $PtSe_2$), we expect that our work will provide a general means to control the catalytic activity of bulk and layered van der Waals crystals that will be critical for other applications including oxygen evolution reactions and CO_2 reduction.

METHODS

Mechanical Exfoliation and Dry Transfer of Layered $PtSe_2$. First, a bulk $PtSe_2$ crystal (CAS: 12038-26-5, supplier: hq$^+$ graphene) was placed on top of Nitto thermal release tape (or base tape). For the mechanical exfoliation of $PtSe_2$, the material was repeatedly stuck and moved (∼100 times) to other areas of the base tape using a second Nitto thermal release tape with a smaller size (i.e., a carrier tape) to form the uniform distribution of $PtSe_2$ flakes with approximately 50% surface coverage. Then, the carrier tape was pressed on the top of the Au/PS substrates with a cotton swab. After adhesion, the prepared sample was heated and pressed with a weight (500 g for the sample area of 5 cm × 5 cm) at 110 °C for 5 min on a hot plate, and then the tape was removed.

Strain Engineering of $PtSe_2$ to Fabricate the Working Electrodes for HER. For the surface wrinkling of $PtSe_2$ electrocatalysts, we used a biaxially prestrained thermoplastic polystyrene (PS) film (Grafix shrink film, Shrink Film, https://www.grafixarts.com/products/shrink-film) which can be shrunk to ∼20% of its original (i.e., prestrained) size by heating the film above its glass transition temperature of 125 °C. This behavior means that we are able to relieve the prestrain (uniaxially or biaxially) by up to $\varepsilon \sim 0.8$.[30,66,67] The PS films were rinsed with ethanol and distilled water following the deposition of Au films (thickness ∼25 nm) using thermal evaporation. For the formation of Au wrinkles and buckled $PtSe_2$ flakes, Au-deposited PS was heated after $PtSe_2$ transfer in a convection oven at 125 °C to relieve prestrain on the PS substrate (Figure S1). To induce the uniaxial compressive strains, we mechanically clamped each end of the PS substrates before heating above the glass transition temperature of PS. We then loaded the mechanically clamped samples in the oven. The PS substrate was heated until the target strain was reached and then cooled to room temperature to stop shrinking. As a result of the strain relief, the PS substrate shrank, and the compressive strain was induced in the $PtSe_2$ flakes and the Au film, forming surface wrinkles. Macroscopic strain (ε) was determined from the change in distance between two lines drawn on the sample before and after the shrinking process using the equation $\varepsilon = (l_0 - l_f)/l_0$, in which l_f is the final length and l_0 is the initial length.[32] To prepare the working electrodes for HER using the strain-engineered $PtSe_2$, we attached copper (Cu) wires to the surfaces of the $PtSe_2$-laden Au films locally and without additional straining by using electrically conductive silver (Ag) paste (Silver paste P-100, CANS). To protect the electrical contact areas on the samples from the electrolyte, we covered the Cu wires with Teflon tubes following the passivation of the contact regions using epoxy resin adhesive (CEMEDINE) (Figure S5).

Surface Topography Measurements Using Atomic Force Microscopy (AFM). We statistically analyzed the thickness and lateral size distribution of as-transferred $PtSe_2$ flakes using atomic force microscopy (MultiMode 8-HR, BRUKER). All the measurements were performed in tapping mode using silicon nitride cantilevers with silicon tips (NCHV-A, BRUKER; resonance frequency of 320 kHz and nominal tip radius of <10 nm).

High-Resolution Transmission Electron Microscopy (HRTEM) to Characterize $PtSe_2$. We performed a TEM analysis to characterize the crystal structures of the mechanically exfoliated $PtSe_2$. First, poly(methyl methacrylate) (PMMA) was spin-coated on $Au/SiO_2/Si$ substrates with transferred $PtSe_2$ flakes at 1500 rpm for 1 min, followed by baking on a hot plate at 120 °C for 5 min. We then separated PMMA/$PtSe_2$ films from the $Au/SiO_2/Si$ substrates by etching the Au films using the gold etchant (TFA, Transene). Upon

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

etching of the Au, the PMMA/$PtSe_2$ films were left floating on the etchant solution. Using clean Si substrates, the PMMA/$PtSe_2$ films were transferred to deionized water to rinse off the gold etchant several times. The resulting films were transferred onto the Quantifoil Cu 1.2/1.3 300 mesh grid (658−300-Cu, Ted Pella) and then baked on a hot plate at 130 °C for 1 h. After baking, the grids were soaked in acetone at 60 °C for 60 min to remove PMMA. The grids were then rinsed with acetone and isopropyl alcohol (IPA) sequentially and then dried on filter paper for 30 min.[68] High-resolution TEM (HRTEM) analysis was carried out using a TEM (TITANTM 80−300, Thermofisher, US) operated at 300 kV. The available point resolution was better than 1 Å at this accelerating voltage. Images of the samples were recorded by a 4k × 4k CCD (Oneview 1095, Gatan, US) camera. The chemical elemental analysis was carried out using a super-X EDS system in a TEM (TALOS F200X, Thermofisher, US) operated at 200 kV, and the available point resolution was better than 1.6 Å at this accelerating voltage. Again, the images of the samples were recorded by a 4k × 4k CCD (Ceta 16M, Thermofisher, US) camera.

X-ray Photoelectron Spectroscopy (XPS) to Characterize $PtSe_2$ before and after Strain Engineering. XPS was performed in a K-α XPS system (Thermo Scientific) with an Al Kα (1487 eV) X-ray source calibrated using the C 1s peak with 400 μm beam spot diameter covering multiple $PtSe_2$ flakes. XPS was used to observe the Pt 4f and Se 3d peaks of the $PtSe_2$ with varying strains, and we analyzed the strain-dependent peak shifts. The XPS valence band spectra were corrected by subtracting the Shirley background. For the accurate comparisons of all valence band spectra before and after strain engineering, the d-band centers of the samples were calculated in the range of −10.0 to 0 eV, using the following equation where $N(E)$ is the DOS, E_d is the center of the d-band, and E is the binding energy.[69,70]

$$E_{\mathrm{d}} = \frac{\int N(E)E\mathrm{d}E}{\int N(E)\mathrm{d}E}$$

Ultraviolet Photoelectron Spectroscopy (UPS) to Characterize $PtSe_2$ before and after Strain Engineering. UPS was performed in a Nexsa UPS system (Thermo Fisher Scientific) with a He I (21.22 eV) source with a 1 mm beam spot diameter (covering multiple $PtSe_2$ flakes) at a base chamber pressure of 2.0×10^{-8} Torr. For measurements, the samples were biased at −10.00 V to measure the work function. The relative valence band maximum of $PtSe_2$ before and after strain engineering was observed at 0 V. A clean gold sample was used to calibrate the Fermi edge of the spectrometer. The mechanically exfoliated $PtSe_2$ on an ITO-coated substrate was used as the pristine sample before strain engineering. To prepare the samples with strained $PtSe_2$, aqueous PVA solution was first spin-coated on the strained $PtSe_2$/Au/PS at 2500 rpm for 1 min. Here, aqueous PVA (poly(vinyl alcohol), DAEJUNG chemical) solution was prepared by dissolving 300 mg of PVA powder in 40 mL deionized water and stirring at 120 °C for 2 h.[71] The samples were then baked on a hot plate at 100 °C for 2 min. Next, PMMA was spin-coated on the PVA/$PtSe_2$/Au/PS at 2500 rpm for 1 min and then baked on a hot plate at 100 °C for 5 min. The PMMA/PVA/$PtSe_2$/Au/PS samples were floated on the surface of gold etchant (TFA, Transene) at 0 °C to etch away the gold film from the substrates. Upon etching of the gold film, PMMA/PVA/$PtSe_2$ films were left floating on the etchant solution. Using clean Si substrates, the films were transferred on the surface of methanol to rinse off the gold etchant. Again, we used the ITO-coated substrate to scoop out the PMMA/PVA/$PtSe_2$ films following the baking on a hot plate (at 130 °C for 60 min). After the hot treatments, the substrate was soaked in deionized water at 130 °C for 120 min to remove PMMA/PVA. Finally, the substrate was rinsed with acetone and IPA and dried on filter paper.

Raman Measurements to Characterize $PtSe_2$ before and after Strain Engineering. Raman spectra of the $PtSe_2$ flakes on Au/PS substrates were obtained using a Raman microscope (UniRAM, UniNanoTech). The measurements were performed in the back-scattering geometry with 532 nm excitation from a diode-pumped solid-state laser. The excitation was focused using a 100× microscope objective with a numerical aperture of 0.9, resulting in a spot diameter of ∼1 μm in which single-flake level characterization is possible. The scattered signal was sent through an 1800 gr/mm grating and collected using a thermoelectrically cooled charge-coupled device (CCD) detector (iDUS DV401A, Andor S5 Technology).

HER Measurements Using Three-Electrode Cells. 0.50 M H_2SO_4 aqueous solution electrolyte was prepared by mixing 1.904 mL of 98% H_2SO_4 (Daejung Chemicals) and distilled water until the total mixture volume was 70.0 mL. We constructed a three-electrode cell configuration in a three-neck flask, using flame-cleaned Pt foil as a counter electrode, 3.0 M Ag/AgCl as a reference electrode (+0.210 V vs RHE), and the $PtSe_2$ catalyst fabricated in this study as the working electrode. We characterized the HER performance of the electrochemical cells containing layered $PtSe_2$ before and after strain engineering using a ZIVE SP1 potentiostat (WonATech). Dry nitrogen (N_2) was bubbled in the electrolyte for 5 min before the electrochemical measurements. During the HER measurements, dry N_2 was passed through the electrolyte with a 1 L/min flow rate, and the dynamic iR correction feature was turned on to monitor and autocorrect the potential-dependent iR drop on a real-time basis. Independent from the geometric area, the C_{DL} measured by CV was used to normalize the current to account for active sites. For the linear sweep voltammetry (LSV) over different overpotential ranges, we set a constant scan rate of 0.2 mV/s.

Calibration of Ag/AgCl Reference Electrode. The calibration of the 3 M Ag/AgCl electrode was performed in a standard three-electrode system. To construct the electrochemical cell for the calibration, Pt foils were used as the working and counter electrodes, and the Ag/AgCl electrodes were used as the reference electrodes. Electrolytes were prepurged and saturated with high-purity H_2. Cyclic voltammetry (CV) was run at a scan rate of 1 mV s^{-1}, and the average potential with j of zero mA/cm^2 during a single cycle of CV was considered as the thermodynamic potential for the hydrogen electrode reaction (Figure S25), referring to the established process.[72]

Electrochemically Active Surface Area (ECSA) Estimation by Cyclic Voltammetry (CV). To characterize the electrochemically active surface area (ECSA), the electrochemical double-layer capacitance (C_{DL}) of the catalytic surfaces in each three-electrode cell (containing $PtSe_2$ strained at different ε) was measured. We characterized C_{DL}, which is linearly proportional to ECSA,[57,58] by measuring the non-Faradaic capacitance current from the scan-rate dependence of CV (Figure S12). Here, we assumed that all the capacitance current is due to double-layer charging as reported from previous literature.[58] To measure double-layer charging via CV, a potential range in which no apparent Faradaic process occurs was determined from static CV, for which the range in this work was centered at the open-circuit potential (OCP) of the system. The C_{DL} was estimated by plotting Δj, the current density difference between the charging and discharging process at OCP, as a function of the scan rate from 20 to 100 mV/s where the slope is twice the value of C_{DL}.[57]

Electrochemical Full-Cell Measurements Using a Lab Scale Two-Electrode Electrolyzer. A two-electrode setup was used to demonstrate the electrochemical full-cell measurements; the $PtSe_2$/Au/PS samples were used as a cathode and IrO_x/CC as an anode (Figure S15). To prepare the IrO_x/CC electrodes, 10 mg of IrO_x powder was first ultrasonically dispersed in a mixture of 965 μL isopropanol and 35 μL of 5 wt % Nafion solution (Sigma-Aldrich) for 1 h. 150 μL of the resulting (homogeneous) ink was then dropped onto both sides of the carbon cloth (CC) with the size of 1×1 cm^2 following the mild drying at 60 °C for 1 h. We used a conventional H-cell configuration to construct the electrolyzer; 30.0 mL of aqueous electrolyte was filled in an individual cell, and a bipolar membrane (FuelCellStore) was utilized to separate the catholyte (0.50 M H_2SO_4) and anolyte (1 M KOH). Before measurements, both electrolytes were purged with dry nitrogen (N_2) for 5 min to eliminate unknown gas species. A ZIVE SP1 potentiostat (WonATech) was used to take the two-electrode full-cell measurements with a scan rate of 5 mV/s.

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

Transfer of Strain-Engineered $PtSe_2$ onto Proton Exchange Membranes (PEMs). The Nafion-coated $PtSe_2$ electrode was prepared by spraying a Nafion ink onto the surface of the electrode. The ink was prepared by mixing 2.5 mL of isopropanol and 30 μL of Nafion (5 wt %) solution. The loading amount of Nafion was set to 12.5 μg cm^{-2}.[73] A commercial HP Nafion membrane (FUELCELL Store, d-170u proton exchange membrane) was placed on a 120 °C hot plate, followed by placing the spray-coated sample to form the composite Nafion membrane with a structured Nafion layer.[74] Finally, the composite PEM system with strained $PtSe_2$ was gently peeled off from the Au/PS substrate (Figure S19).

Measurements of Water Contact Angle to Characterize the Surface Wetting Properties of Samples. To characterize the surface wettability of Au/PS surfaces, we measured the static water contact angle (θ_W), advancing contact angle (θ_A), and receding contact angle (θ_R) at room temperature using a surface analyzer (DROP SHAPE ANALYZER−DSA25, KRÜSS, DE). For every measurement, we used an initial water droplet size of 6 μL, and at least ten measurements for each sample were averaged. To calculate contact angle hysteresis (θ_H), the measured θ_R was subtracted from the measured θ_A.[75]

Measurement of the Gaseous Product in Real Time to Calculate Faradaic Efficiency. To measure the Faradaic efficiency, an Agilent GC 8890A with a thermal conductivity detector was used. The GC system was calibrated using a certified standard of hydrogen before use (Shinyang Medicine Co. Ltd., Korea). The faradaic efficiency measurements were performed using an airtight electrochemical cell purged with Ar for 30 min. We used a well-established method from the published literature to calculate the theoretical amount of H_2 for the given experimental conditions.[76]

Density Functional Theory (DFT) Calculation. Density functional theory (DFT) calculations were performed by using the Vienna ab initio Software Package (VASP) 5.4.4.[77] The Perdew−Burke−Ernzerhof (PBE) exchange-correlation functional was employed with Grimme's D3 method for dispersion correction.[78,79] The pseudopotentials were obtained from the projector-augmented wave (PAW) method,[80] and the kinetic energy cutoff for plane waves was set to 700 eV for the calculation of electronic structure in Figure 3 and 400 eV for the calculation of the edge-plane model in Figure 5. The band structure and density of states in Figure 3 were obtained by spin−orbit coupling calculation of the orthorhombic trilayer unit cell with a 20 Å vacuum on the basal plane. The edge plane was modeled by trilayer, each of which has 4 PtSe2 units, and the vacuum was set as 15 Å thick. The geometry and charge density of atomic structures were visualized by VESTA 3.5.8.[81] The geometry was optimized until the maximum force became less than 0.01 eV/Å, and only the H atom was relaxed for the slab calculations to constrain the assigned adsorption configurations. For the theoretical comparison, the adsorption free energy change of an H atom on the Pt(111) surface was calculated by a Pt(111) slab model of a 2 × 8 supercell with four atomic layers (Figure S20), having a comparable surface area with the edge-plane model of $PtSe_2$ in the main text (Figure 5a), and the bottom two layers were fixed during optimization. The free energy of adsorption was calculated by the entropy correction, $\Delta G_H = \Delta E_H + 0.24$ eV.[62]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsnano.4c18077.

Fabrication of strain-engineered $PtSe_2$ catalysts for HER; crystal structure of layered $PtSe_2$; Raman spectra of strained $PtSe_2$ before and after HER measurement; work functions analysis of $PtSe_2$ before and after strain engineering; formation of $PtSe_2$/Au/PS working electrode for HER; HER performance of as-transferred $PtSe_2$ compared with Au films; electrochemical characterization of strained $PtSe_2$ before and after iR correction; EIS measurements of $PtSe_2$ before and after strain engineering; surface wetting properties of Au/PS surface before and after strain engineering; HER activity of Au/PS substrate before and after strain engineering; HER performance of $PtSe_2$ engineered at system maximum strain; measurements of non-Faradaic capacitance current to estimate C_{DL}; extrapolation method to obtain j_0 from Tafel plots; chronoamperometry measurements of strained $PtSe_2$ and Pt; schematic illustration of a two-electrode electrolyzer system; OER performance of IrO_x/CC electrode; chronoamperometry measurement of strained $PtSe_2$ in electrolyzer; HER performance of strained $PtSe_2$ in PEM; nanoimprinting transfer processes of strained $PtSe_2$ onto PEM for water electrolysis; Pt(111) 2 × 8 supercell model; dihedral angles of active sites near edge planes with- and without *H intermediates; extended electronic state analysis of $PtSe_2$ with compressive and tensile strains; transition from intra- to interlayer electron coupling with *H intermediates under varying strain; XPS survey spectrum of $PtSe_2$ before and after strain engineering; calibration of Ag/AgCl reference electrode; comparisons of HER performance of Pt-based TMDs electrocatalyst, and supporting references (PDF)

AUTHOR INFORMATION

Corresponding Authors

In Soo Kim − *Nanophotonics Research Center, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea; KIST-SKKU Carbon-Neutral Research Center, Sungkyunkwan University (SKKU), Suwon 16419, Republic of Korea;* orcid.org/0000-0003-1615-3119; Email: isk@kist.re.kr

Minho Kim − *Department of Applied Chemistry, Kyung Hee University, Yongin 17104, Republic of Korea;* Email: minho.kim@khu.ac.kr

Won-Kyu Lee − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea;* orcid.org/0000-0002-7878-1981; Email: wklee@hongik.ac.kr

Authors

Hotae Jeon − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*

Hee Jung Kwon − *Department of Applied Chemistry, Kyung Hee University, Yongin 17104, Republic of Korea*

Jaehyun Lee − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea;* orcid.org/0009-0003-7468-7319

Sun Kyung Han − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*

Hyunjin Kim − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*

Jaewon Heo − *Nanophotonics Research Center, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea*

Junhwi Han − *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

Seunghun Shin – *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*
Jiheon Park – *Department of Materials Science and Engineering, Hongik University, Seoul 04066, Republic of Korea*
Min Kyung Cho – *Advanced Analysis and Data Center, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea*
Daniel J. Preston – *Department of Mechanical Engineering, Rice University, Houston, Texas 77005, United States; Rice Advanced Materials Institute (RAMI), Rice University, Houston, Texas 77005, United States;* orcid.org/0000-0002-0096-0285

Complete contact information is available at:
https://pubs.acs.org/10.1021/acsnano.4c18077

Author Contributions

○H.J., H.J.K., and J.L. contributed equally to this work. W.K.L. developed the concept and designed the experiments. H.J. and J.L. fabricated samples and prepared the electrochemical cells to test HER performance. H.J., S.K.H., and W.K.L. characterized the samples using AFM and SEM. H.J., J.L., and I.S.K. measured UPS and Raman spectroscopy. H.J., J.L., and J.H. measured XPS spectroscopy. H.J., J.L., J.P., H.S., and M.K.C. characterized the samples using TEM. H.J.K. and M.K. developed theoretical models. D.J.P., I.S.K., M.K., and W.K.L. analyzed experimental and theoretical data. W.K.L. supervised the experiments. H.J., H.J.K., J.L., D.J.P., I.S.K., M.K., and W.K.L. wrote manuscripts.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (No. 2022R1C1C1010434). This work was also supported by Korea Basic Science Institute (National Research Facilities and Equipment Center) grant funded by the Ministry of Science and ICT (No. RS-2024-00402372). This work was partially supported by 2023 Hongik University Research Fund. H.J.K. and M.K. were also supported by the National Supercomputing Center with the supercomputing resources including technical support (KSC-2023-CRE-0558). I.S.K. acknowledges the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (RS-2024-00358623) and KIST intramural grants (2V10250 and 2E33221). D.J.P. acknowledges internal funding support from Rice University.

REFERENCES

(1) Zhao, Y.; Qiao, J.; Yu, Z.; Yu, P.; Xu, K.; Lau, S. P.; Zhou, W.; Liu, Z.; Wang, X.; Ji, W.; Chai, Y. High-Electron-Mobility and Air-Stable 2D Layered $PtSe_2$ FETs. *Adv. Mater.* **2017**, *29* (5), No. 1604230.
(2) Gong, Y.; Lin, Z.; Chen, Y.-X.; Khan, Q.; Wang, C.; Zhang, B.; Nie, G.; Xie, N.; Li, D. Two-Dimensional Platinum Diselenide: Synthesis, Emerging Applications, and Future Challenges. *Nano-Micro Lett.* **2020**, *12* (1), No. 174.
(3) Pi, L.; Li, L.; Liu, K.; Zhang, Q.; Li, H.; Zhai, T. Recent Progress on 2D Noble-Transition-Metal Dichalcogenides. *Adv. Funct. Mater.* **2019**, *29* (51), No. 1904932.
(4) Wang, Z.; Li, Q.; Besenbacher, F.; Dong, M. Facile Synthesis of Single Crystal $PtSe_2$ Nanosheets for Nanoscale Electronics. *Adv. Mater.* **2016**, *28* (46), 10224–10229.
(5) Ansari, L.; Monaghan, S.; McEvoy, N.; Coileáin, C. Ó.; Cullen, C. P.; Lin, J.; Siris, R.; Stimpel-Lindner, T.; Burke, K. F.; Mirabelli, G.; Duffy, R.; Caruso, E.; Nagle, R. E.; Duesberg, G. S.; Hurley, P. K.; Gity, F. Quantum confinement-induced semimetal-to-semiconductor evolution in large-area ultra-thin $PtSe_2$ films grown at 400 °C. *npj 2D Mater. Appl.* **2019**, *3* (33), No. 33, DOI: 10.1038/s41699-019-0116-4.
(6) Almutairi, A.; Yin, D.; Yoon, Y. $PtSe_2$ Field-Effect Transistors: New Opportunities for Electronic Devices. *IEEE Electron Device Lett.* **2018**, *39* (1), 151–154.
(7) Kim, Y.-H.; Kang, M.-S.; Choi, J. W.; Lee, W.-Y.; Kim, M.-J.; Park, N.-W.; Yoon, Y.-G.; Kim, G.-S.; Lee, S.-K. Barrier-free semimetallic $PtSe_2$ contact formation in two-dimensional $PtSe_2$/$PtSe_2$ homostructure for high-performance field-effect transistors. *Appl. Surf. Sci.* **2023**, *638*, No. 158061.
(8) Wang, G. Z.; Wang, Z. Z.; McEvoy, N.; Fan, P.; Blau, W. J. Layered $PtSe_2$ for Sensing, Photonic, and (Opto-)Electronic Applications. *Adv. Mater.* **2021**, *33* (1), No. 2004070.
(9) Wang, Y.; Li, L.; Yao, W.; Song, S.; Sun, J. T.; Pan, J.; Ren, X.; Li, C.; Okunishi, E.; Wang, Y.-Q.; Wang, E.; Shao, Y.; Zhang, Y. Y.; Yang, H.-T.; Schwier, E. F.; Iwasawa, H.; Shimada, K.; Taniguchi, M.; Cheng, Z.; Zhou, S.; Du, S.; Pennycook, S. J.; Pantelides, S. T.; Gao, H.-J. Monolayer $PtSe_2$, a New Semiconducting Transition-Metal-Dichalcogenide, Epitaxially Grown by Direct Selenization of Pt. *Nano Lett.* **2015**, *15* (6), 4013–4018.
(10) Yim, C.; Lee, K.; McEvoy, N.; O'Brien, M.; Riazimehr, S.; Berner, N. C.; Cullen, C. P.; Kotakoski, J.; Meyer, J. C.; Lemme, M. C.; Duesberg, G. S. High-Performance Hybrid Electronic Devices from Layered $PtSe_2$ Films Grown at Low Temperature. *ACS Nano* **2016**, *10*, 9550–9558.
(11) Wang, G.; Wang, K.; McEvoy, N.; Bai, Z.; Cullen, C. P.; Murphy, C. N.; McManus, J. B.; Magan, J. J.; Smith, C. M.; Duesberg, G. S.; Kaminer, I.; Wang, J.; Blau, W. J. Ultrafast Carrier Dynamics and Bandgap Renormalization in Layered $PtSe_2$. *Small* **2019**, *15* (34), No. 1902728.
(12) Wang, P.; He, D.; Wang, Y.; Zhang, X.; He, J.; Zhao, H. Fast Exciton Diffusion in Monolayer $PtSe_2$. *Laser Photonics Rev.* **2022**, *16* (7), No. 2100594.
(13) Abdukayumov, K.; Micica, M.; Ibrahim, F.; Vojácek, L.; Vergnaud, C.; Marty, A.; Veuillen, J.-Y.; Mallet, P.; de Moraes, I. G.; Dosenovic, D.; Gambarelli, S.; Maurel, V.; Wright, A.; Tignon, J.; Mangeney, J.; Ouerghi, A.; Renard, V.; Mesple, F.; Li, J.; Bonell, F.; Okuno, H.; Chshiev, M.; George, J.-M.; Jaffres, H.; Dhillon, S.; Jamet, M. Atomic-Layer Controlled Transition from Inverse Rashba–Edelstein Effect to Inverse Spin Hall Effect in 2D $PtSe_2$ Probed by THz Spintronic Emission. *Adv. Mater.* **2024**, *36* (14), No. 2304243.
(14) Hemmat, M.; Ayari, S.; Mičica, M.; Vergnet, H.; Guo, S.; Arfaoui, M.; Yu, X.; Vala, D.; Wright, A.; Postava, K.; Mangeney, J.; Carosella, F.; Jaziri, S.; Wang, Q. J.; Liu, Z.; Tignon, J.; Ferreira, R.; Baudin, E.; Dhillon, S. Layer-controlled nonlinear terahertz valleytronics in two-dimensional semimetal and semiconductor $PtSe_2$. *InfoMat* **2023**, *5* (11), No. e12468.
(15) Yao, W.; Wang, E.; Huang, H.; Deng, K.; Yan, M.; Zhang, K.; Miyamoto, K.; Okuda, T.; Li, L.; Wang, Y.; Gao, H.; Liu, C.; Duan, W.; Zhou, S. Direct observation of spin-layer locking by local Rashba effect in monolayer semiconducting $PtSe_2$ film. *Nat. Commun.* **2017**, *8* (1), No. 14216.
(16) Villaos, R. A. B.; Crisostomo, C. P.; Huang, Z. Q.; Huang, S. M.; Padama, A. A. B.; Albao, M. A.; Lin, H.; Chuang, F. C. Thickness dependent electronic properties of Pt dichalcogenides. *npj 2D Mater. Appl.* **2019**, *3*, No. 2.
(17) Hu, D. K.; Zhao, T. Q.; Ping, X. F.; Zheng, H. S.; Xing, L.; Liu, X. Z.; Zheng, J. Y.; Sun, L. F.; Gu, L.; Tao, C. G.; Wang, D.; Jiao, L. Unveiling the Layer-Dependent Catalytic Activity of $PtSe_2$ Atomic Crystals for the Hydrogen Evolution Reaction. *Angew. Chem., Int. Ed.* **2019**, *58* (21), 6977–6981.

(18) He, Y.; Liu, L.; Zhu, C.; Guo, S.; Golani, P.; Koo, B.; Tang, P.; Zhao, Z.; Xu, M.; Zhu, C.; Yu, P.; Zhou, X.; Gao, C.; Wang, X.; Shi, Z.; Zheng, L.; Yang, J.; Shin, B.; Arbiol, J.; Duan, H.; Du, Y.; Heggen, M.; Dunin-Borkowski, R. E.; Guo, W.; Wang, Q. J.; Zhang, Z.; Liu, Z. Amorphizing noble metal chalcogenide catalysts at the single-layer limit towards hydrogen production. *Nat. Catal.* **2022**, *5*, 212−221.
(19) Wang, Z.; Xiao, B.; Lin, Z.; Xu, Y.; Lin, Y.; Meng, F.; Zhang, Q.; Gu, L.; Fang, B.; Guo, S.; Zhong, W. $PtSe_2$/Pt Heterointerface with Reduced Coordination for Boosted Hydrogen Evolution Reaction. *Angew. Chem., Int. Ed.* **2021**, *60* (43), 23388−23393.
(20) Chen, K.; Huan, Y.; Quan, W.; Zhu, L.; Fu, J.; Hu, J.; Cui, F.; Zhou, F.; Wang, X.; Li, M.; Zhang, Y. Controllable Growth and Defect Engineering of Vertical $PtSe_2$ Nanosheets for Electrocatalytic Hydrogen Evolution. *ACS Energy Lett.* **2022**, *7*, 3675−3684.
(21) Deng, S.; Sumant, A. V.; Berry, V. Strain engineering in two-dimensional nanomaterials beyond graphene. *Nano Today* **2018**, *22*, No. 14.
(22) Yan, Y.; Ding, S.; Wu, X.; Zhu, J.; Feng, D.; Yang, X.; Li, F. Tuning the physical properties of ultrathin transition-metal dichalcogenides via strain engineering. *RSC Adv.* **2020**, *10* (65), 39455−39467.
(23) Wu, W.; Wang, J.; Ercius, P.; Wright, N. C.; Leppert-Simenauer, D. M.; Burke, R. A.; Dubey, M.; Dogare, A. M.; Pettes, M. T. Giant Mechano-Optoelectronic Effect in an Atomically Thin Semiconductor. *Nano Lett.* **2018**, *18*, 2351−2357.
(24) Dhakal, K. P.; Roy, S.; Jang, H.; Chen, X.; Yun, W. S.; Kim, H.; Lee, J.; Kim, J.; Ahn, J.-H. Local Strain Induced Band Gap Modulation and Photoluminescence Enhancement of Multilayer Transition Metal Dichalcogenides. *Chem. Mater.* **2017**, *29*, No. 12.
(25) Zhao, X.; Tang, G.; Li, Y.; Zhang, M.; Nie, Y. Biaxial Strain Improving the Thermoelectric Performance of a Two-Dimensional MoS_2/WS_2 Heterostructure. *ACS Appl. Electron. Mater.* **2021**, *3* (7), 2995−3004.
(26) Tangpakonsab, P.; Moontragoon, P.; Hussain, T.; Kaewmaraya, T. Thermoelectric Efficiency of Two-Dimensional Pentagonal-$PdSe_2$ at High Temperatures and the Role of Strain. *ACS Appl. Energy Mater.* **2022**, *5* (11), 14522−14530.
(27) Castellanos-Gomez, A.; Roldán, R.; Cappelluti, E.; Buscema, M.; Guinea, F.; van der Zant, H. S. J.; Steele, G. A. Local Strain Engineering in Atomically Thin MoS_2. *Nano Lett.* **2013**, *13* (11), 5361−5366.
(28) Cho, C.; Wong, J.; Taqieddin, A.; Biswas, S.; Aluru, N. R.; Nam, S.; Atwater, H. A. Highly Strain-Tunable Interlayer Excitons in MoS_2/WSe_2 Heterobilayers. *Nano Lett.* **2021**, *21* (9), 3956−3964.
(29) Rhuy, D.; Lee, Y.; Kim, J. Y.; Kim, C.; Kwon, Y.; Preston, D. J.; Kim, I. S.; Odom, T. W.; Kang, K.; Lee, D.; Lee, W.-K. Ultraefficient Electrocatalytic Hydrogen Evolution from Strain-Engineered, Multilayer MoS_2. *Nano Lett.* **2022**, *22* (14), 5742−5750.
(30) Lee, Y.; Lee, S. H.; Han, S. K.; Park, J.; Lee, D.; Preston, D. J.; Kim, I. S.; Hersam, M. C.; Kwon, Y.; Shong, B.; Lee, W.-K. Strain-Enabled Local Phase Control in Layered $MoTe_2$ for Enhanced Electrocatalytic Hydrogen Evolution. *ACS Energy Lett.* **2023**, *8* (11), 4716−4725.
(31) Li, P.; Li, L.; Zeng, X. C. Tuning the electronic properties of monolayer and bilayer $PtSe_2$ via strain engineering. *J. Mater. Chem. C* **2016**, *4* (15), 3106−3112.
(32) Lee, W.-K.; Kang, J.; Chen, K.-S.; Engel, C. J.; Jung, W.-B.; Rhee, D.; Hersam, M. C.; Odom, T. W. Multiscale, Hierarchical Patterning of Graphene by Conformal Wrinkling. *Nano Lett.* **2016**, *16* (11), 7121−7127.
(33) Lee, W.-K.; Odom, T. W. Designing Hierarchical Nanostructures from Conformable and Deformable Thin Materials. *ACS Nano* **2019**, *13* (6), 6170−6177.
(34) Rhee, D.; Paci, J. T.; Deng, S.; Lee, W.-K.; Schatz, G. C.; Odom, T. W. Soft Skin Layers Enable Area-Specific, Multiscale Graphene Wrinkles with Switchable Orientations. *ACS Nano* **2020**, *14* (1), 166−174.
(35) Wang, M. C.; Chun, S.; Han, R. S.; Ashraf, A.; Kang, P.; Nam, S. Heterogeneous, Three-Dimensional Texturing of Graphene. *Nano Lett.* **2015**, *15* (3), 1829−1835.
(36) Lee, W.-K.; Jung, W.-B.; Rhee, D.; Hu, J.; Lee, Y.-A. L.; Jacobson, C.; Jung, H.-T.; Odom, T. W. Monolithic Polymer Nanoridges with Programmable Wetting Transitions. *Adv. Mater.* **2018**, *30* (32), No. 1706657.
(37) El Sachat, A.; Xiao, P.; Donadio, D.; Bonell, F.; Sledzinska, M.; Marty, A.; Vergnaud, C.; Boukari, H.; Jamet, M.; Arregui, G.; Chen, Z.; Alzina, F.; Torres, C. M. S.; Chavez-Angel, E. Effect of crystallinity and thickness on thermal transport in layered $PtSe_2$. *npj 2D Mater. Appl.* **2022**, *6*, No. 32.
(38) Li, K.; Wang, T.; Wang, W.; Gao, X. Lattice vibration properties of $MoS_2/PtSe_2$ heterostructures. *J. Alloys Compd.* **2020**, *820*, No. 153192.
(39) Gulo, D. P.; Yeh, H.; Chang, W.-H.; Liu, H.-L. Temperature-dependent optical and vibrational properties of $PtSe_2$ thin films. *Sci. Rep.* **2020**, *10*, No. 19003.
(40) Conley, H. J.; Wang, B.; Ziegler, J. I.; Haglund, R. F., Jr; Pantelides, S. T.; Bolotin, K. I. Bandgap Engineering of Strained Monolayer and Bilayer MoS_2. *Nano Lett.* **2013**, *13*, 3626−3630.
(41) Lee, F.; Tripathi, M.; Sanchez Salas, R.; Ogilvie, S. P.; Amorim Graf, A.; Jurewicz, I.; Dalton, A. B. Localised strain and doping of 2D materials. *Nanoscale* **2023**, *15*, 7227−7248.
(42) Youn, S.; Kim, J.; Moon, H.; Kim, J.; Jang, J.; Chang, J.; Lee, T.; Kang, K.; Lee, W. Enhanced Thermoelectric Power Factor in Carrier-Type-Controlled Platinum Diselenide Nanosheets by Molecular Charge-Transfer Doping. *Small* **2022**, *18* (23), No. 2200818.
(43) Withanage, S. S.; Chamlagain, B.; Johnston, A. C.; Khondaker, S. I. Charge Transfer Doping of 2D $PdSe_2$ Thin Film and Its Application in Fabrication of Heterostructures. *Adv. Electron. Mater.* **2021**, *7* (3), No. 2001057.
(44) Liu, K.; Zheng, B. J.; Wu, J. J.; Chen, Y. F.; Wang, X. Q.; Qi, F.; He, D. W.; Zhang, W. L.; Li, Y. R. Synthesis of two-dimensional semiconductor single-crystal $PtSe_2$ under high pressure. *J. Mater. Sci.* **2018**, *53* (2), 1256−1263.
(45) Mirabelli, G.; Walsh, L. A.; Gity, F.; Bhattacharjee, S.; Cullen, C. P.; Coileáin, C. Ó.; Monaghan, S.; McEvoy, N.; Nagle, R.; Hurley, P. K.; Duffy, R. Effects of Annealing Temperature and Ambient on Metal/$PtSe_2$ Contact Alloy Formation. *ACS Omega* **2019**, *4* (17), 17487−17493.
(46) Zeng, L.; Lin, S.; Lou, Z.; Yuan, H.; Long, H.; Li, Y.; Lu, W.; Lau, S. P.; Wu, D.; Tsang, Y. H. Ultrafast and sensitive photodetector based on a $PtSe_2$/silicon nanowire array heterojunction with a multiband spectral response from 200 to 1550 nm. *NPG Asia Mater.* **2018**, *10* (4), 352−362.
(47) Chung, C. C.; Yeh, H.; Wu, P. H.; Lin, C. C.; Li, C. S.; Yeh, T. T.; Chou, Y.; Wei, C. Y.; Wen, C. Y.; Chou, Y. C.; Luo, C. W.; Wu, C. I.; Li, M. Y.; Li, L. J.; Chang, W. H.; Chen, C. W. Atomic-Layer Controlled Interfacial Band Engineering at Two-Dimensional Layered $PtSe_2$/Si Heterojunctions for Efficient Photoelectrochemical Hydrogen Production. *ACS Nano* **2021**, *15* (3), 4627−4635.
(48) Fan, J.-L.; Qin, W.-W.; Liu, Z.-Y.; Zhou, M. One-step synthesis of two-dimensional $PdSe_2/PtSe_2$ heterostructures. *Mater. Lett.* **2023**, *352*, No. 135210.
(49) Lanzillo, N. A.; Simbeck, A. J.; Nayak, S. K. Strain engineering the work function in monolayer metal dichalcogenides. *J. Phys.: Condens. Matter* **2015**, *27* (17), No. 175501.
(50) Luo, M.; Guo, S. Strain-controlled electrocatalysis on multimetallic nanomaterials. *Nat. Rev. Mater.* **2017**, *2*, No. 17059.
(51) Hammer, B.; Nørskov, J. K. Electronic factors determining the reactivity of metal surfaces. *Surf. Sci.* **1995**, *343* (3), 211−220.
(52) Wang, Z.; Shen, S.; Wang, J.; Zhong, W. Modulating the D-Band Center of Electrocatalysts for Enhanced Water Splitting. *Chem. - Eur. J.* **2024**, *30*, No. e202402725.
(53) Yu, Y.; Huang, S. Y.; Li, Y.; Steinmann, S. N.; Yang, W.; Cao, L. Layer-Dependent Electrocatalysis of MoS_2 for Hydrogen Evolution. *Nano Lett.* **2014**, *14* (2), 553−558.

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120

(54) Lin, J. W.; Chen, C. W.; Chen, P. T. Interlayer Interaction Induced Layer-Dependent Catalytic Activity toward a Hydrogen Evolution Reaction on Two-Dimensional $PtSe_2$. *J. Phys. Chem. C* **2021**, *125* (36), 19716−19723.

(55) Alinezhad, A.; Benedetti, T. M.; Lian, J.; Gonçales, V. R.; Gooding, J. J.; Tilley, R. D. Controlling hydrogen evolution reaction activity on Ni core−Pt island nanoparticles by tuning the size of the Pt islands. *Chem. Commun.* **2021**, *57* (22), 2788−2791.

(56) Yu, F.-Y.; Lang, Z.-L.; Yin, L.-Y.; Feng, K.; Xia, Y.-J.; Tan, H.-Q.; Zhu, H.-T.; Zhong, J.; Kang, Z.-H.; Li, Y.-G. Pt-O bond as an active site superior to Pt^0 in hydrogen evolution reaction. *Nat. Commun.* **2020**, *11*, No. 490.

(57) Lukowski, M. A.; Daniel, A. S.; Meng, F.; Forticaux, A.; Li, L.; Jin, S. Enhanced Hydrogen Evolution Catalysis from Chemically Exfoliated Metallic MoS_2 Nanosheets. *J. Am. Chem. Soc.* **2013**, *135* (28), 10274−10277.

(58) McCrory, C. C. L.; Jung, S.; Peters, J. C.; Jaramillo, T. F. Benchmarking Heterogeneous Electrocatalysts for the Oxygen Evolution Reaction. *J. Am. Chem. Soc.* **2013**, *135* (45), 16977−16987.

(59) Li, X.; Fang, Y.; Wang, J.; Fang, H.; Xi, S.; Zhao, X.; Xu, D.; Xu, H.; Yu, W.; Hai, X.; Chen, C.; Yao, C.; Tao, H. B.; Howe, A. G. R.; Pennycook, S. J.; Liu, B.; Lu, J.; Su, C. Ordered clustering of single atomic Te vacancies in atomically thin $PtTe_2$ promotes hydrogen evolution catalysis. *Nat. Commun.* **2021**, *12*, No. 2351.

(60) Li, X.; Wang, Y.; Wang, J.; Da, Y.; Zhang, J.; Li, L.; Zhong, C.; Deng, Y.; Han, X.; Hu, W. Sequential Electrodeposition of Bifunctional Catalytically Active Structures in MoO_3/Ni−NiO Composite Electrocatalysts for Selective Hydrogen and Oxygen Evolution. *Adv. Mater.* **2020**, *32*, No. 2003414.

(61) Zhu, J.; Wang, Z.-C.; Dai, H.; Wang, Q.; Yang, R.; Yu, H.; Liao, M.; Zhang, J.; Chen, W.; Wei, Z.; Li, N.; Du, L.; Shi, D.; Wang, W.; Zhang, L.; Jiang, Y.; Zhang, G. Boundary activated hydrogen evolution reaction on monolayer MoS_2. *Nat. Commun.* **2019**, *10*, No. 1348.

(62) Nørskov, J. K.; Bligaard, T.; Logadottir, A.; Kitchin, J. R.; Chen, J. G.; Pandelov, S.; Stimming, U. Trends in the Exchange Current for Hydrogen Evolution. *J. Electrochem. Soc.* **2005**, *152* (3), No. J23.

(63) Gregory, B. Z.; Strempfer, J.; Weinstock, D.; Ruf, J. P.; Sun, Y.; Nair, H.; Schreiber, N. J.; Schlom, D. G.; Shen, K. M.; Singer, A. Strain-induced orbital-energy shift in antiferromagnetic RuO_2 revealed by resonant elastic x-ray scattering. *Phys. Rev. B* **2022**, *106* (19), No. 195135.

(64) Xu, Y.; Li, Y.; Chen, X.; Zhang, C.; Zhang, R.; Lu, P. First-principle study of hydrogenation on monolayer MoS_2. *AIP Adv.* **2016**, *6*, No. 075001.

(65) Hu, T.; Li, R.; Dong, J. A new (2×1) dimerized structure of monolayer 1T-molybdenum disulfide, studied from first principles calculations. *J. Chem. Phys.* **2013**, *139*, No. 174702.

(66) Lee, W.-K.; Engel, C. J.; Huntington, M. D.; Hu, J.; Odom, T. W. Controlled Three-Dimensional Hierarchical Structuring by Memory-Based, Sequential Wrinkling. *Nano Lett.* **2015**, *15* (8), 5624−5629.

(67) Huntington, M. D.; Engel, C. J.; Odom, T. W. Controlling the Orientation of Nanowrinkles and Nanofolds by Patterning Strain in a Thin Skin Layer on a Polymer Substrate. *Angew. Chem., Int. Ed.* **2014**, *53* (31), 8117−8121.

(68) Park, J.; Cho, I.; Jeon, H.; Lee, Y.; Zhang, J.; Lee, D.; Cho, M. K.; Preston, D. J.; Shong, B.; Kim, I. S.; Lee, W.-K. Conversion of Layered WS_2 Crystals into Mixed-Domain Electrochemical Catalysts by Plasma-Assisted Surface Reconstruction. *Adv. Mater.* **2024**, *36* (25), No. 2314031.

(69) Song, Q.; Li, J.; Wang, S.; Liu, J.; Liu, X.; Pang, L.; Li, H.; Liu, H. Enhanced Electrocatalytic Performance through Body Enrichment of Co-Based Bimetallic Nanoparticles In Situ Embedded Porous N-Doped Carbon Spheres. *Small* **2019**, *15* (44), No. 1903395.

(70) Tao, L.; Wang, K.; Lv, F.; Mi, H.; Lin, F.; Luo, H.; Guo, H.; Zhang, Q.; Gu, L.; Luo, M.; Guo, S. Precise Synthetic Control of Exclusive Ligand Effect Boosts Oxygen Reduction Catalysis. *Nat. Commun.* **2023**, *14*, No. 6893.

(71) Van Ngoc, H.; Qian, Y.; Han, S. K.; Kang, D. J. PMMA-Etching-Free Transfer of Wafer-Scale Chemical Vapor Deposition Two-Dimensional Atomic Crystal by a Water Soluble Polyvinyl Alcohol Polymer Method. *Sci. Rep.* **2016**, *6*, No. 33096.

(72) Zhu, Y.; Zhou, W.; Zhong, Y.; Bu, Y.; Chen, X.; Zhong, Q.; Liu, M.; Shao, Z. A Perovskite Nanorod as Bifunctional Electrocatalyst for Overall Water Splitting. *Adv. Energy Mater.* **2017**, *7* (8), No. 1602122.

(73) Shin, S. J.; Choi, H.; Ringe, S.; Won, D. H.; Oh, H. S.; Kim, D. H.; Lee, T.; Nam, D. H.; Kim, H.; Choi, C. H. A Unifying Mechanism for Cation Effect Modulating C1 and C2 Productions from CO2 Electroreduction. *Nat. Commun.* **2022**, *13*, No. 5482.

(74) Li, Y.; Wen, Q.; Qin, J.; Zou, S.; Ning, F.; Bai, C.; Pan, S.; Jin, H.; Xu, P.; Shen, M.; Song, Y. J.; Zhou, X. A High-Efficient and Low-Consumption Nanoimprint Method to Prepare Large-Area and High-Quality Nafion Array for the Ordered MEA of Fuel Cell. *Chem. Eng. J.* **2023**, *451*, No. 138722.

(75) Lee, W.-K.; Jung, W.-B.; Nagel, S. R.; Odom, T. W. Stretchable Superhydrophobicity from Monolithic, Three-Dimensional Hierarchical Wrinkles. *Nano Lett.* **2016**, *16* (6), 3774−3779.

(76) Wu, H.; Song, J.; Xie, C.; Hu, Y.; Zhang, P.; Yang, G.; Han, B. Surface Engineering in PbS via Partial Oxidation: Towards an Advanced Electrocatalyst for Reduction of Levulinic Acid to γ-Valerolactone. *Chem. Sci.* **2019**, *10* (6), 1754−1759.

(77) Kresse, G.; Hafner, J. *Ab Initio* Molecular-Dynamics Simulation of the Liquid-Metal−Amorphous-Semiconductor Transition in Germanium. *Phys. Rev. B* **1994**, *49* (20), No. 14251.

(78) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. *Phys. Rev. Lett.* **1996**, *77*, No. 3865.

(79) Grimme, S.; Ehrlich, S.; Goerigk, L. Effect of the Damping Function in Dispersion Corrected Density Functional Theory. *J. Comput. Chem.* **2011**, *32* (7), 1456−1465.

(80) Kresse, G.; Joubert, D. From Ultrasoft Pseudopotentials to the Projector Augmented-Wave Method. *Phys. Rev. B* **1999**, *59* (3), No. 1758.

(81) Momma, K.; Izumi, F. VESTA 3 for Three-Dimensional Visualization of Crystal, Volumetric and Morphology Data. *J. Appl. Crystallogr.* **2011**, *44* (6), 1272−1276.

https://doi.org/10.1021/acsnano.4c18077
ACS Nano 2025, 19, 9107−9120